

ResearchForge Training OS System Design & Workflow

Vision

Build a CLI-first operating system for AI/ML research and LLM training. The platform orchestrates research discovery, dataset validation, experiment planning, training, evaluation, and continuous optimization while minimizing hallucinations by keeping deterministic systems responsible for data retrieval, validation, and execution.

Core Principle

Evidence → Knowledge Graph → AI Reasoning → Verification → Execution.
LLMs never directly decide factual information that can be retrieved or measured.

High-Level Architecture

1. CLI (researchforge)
2. Intent Parser (Gemini): extract structured JSON (task, modality, metrics, hardware).
3. Parallel Search: ArXiv, Semantic Scholar, OpenAlex, GitHub, PapersWithCode, Google Dataset Search, Hugging Face, Kaggle, OpenML.
4. Evidence Store (PostgreSQL + object storage).
5. Knowledge Graph (Neo4j): papers, datasets, models, metrics, tasks, authors.
6. Reasoning Layer (Gemini): synthesize evidence, recommend datasets/models with justification.
7. Dataset Validation: Great Expectations, Pandera, Cleanlab, Polars/DuckDB.
8. Training Planner (rule engine): choose pipeline/templates.
9. Config Generator (YAML/JSON).
10. Execution Engine: PyTorch Lightning, Accelerate, Transformers, TRL.
11. Experiment Manager: Optuna / Ray Tune.
12. Evaluation: statistical significance + benchmark comparison.
13. AutoResearch Agent: literature monitoring, experiment proposals, reproducible reports.

Knowledge Graph Schema

Nodes: Paper, Dataset, Model, Task, Metric, Benchmark, Repository, Author.
Edges: uses_dataset, evaluates_on, beats, extends, cites, implements, trained_on.

Hallucination Prevention

- Search via APIs, not LLM generation.
- Dataset ranking from measurable metadata.
- Cleaning via deterministic libraries.
- Templates/config generation instead of raw code generation.
- Metrics determine success; LLM summarizes results.

CLI Commands

```
researchforge init
researchforge research
researchforge datasets
researchforge validate
researchforge plan
researchforge train
```

researchforge tune
researchforge evaluate
researchforge report
researchforge autoresearch

End-to-End Workflow

User Prompt

- Intent Extraction
- Parallel Evidence Search
- Knowledge Graph Construction
- Dataset Discovery & Validation
- Training Plan
- Config Generation
- Execution
- Hyperparameter Optimization
- Evaluation
- AutoResearch Loop
- Final Report & Deployment

Future Extensions

GPU provider abstraction (RunPod, Modal, Lambda, local), multi-agent orchestration, checkpoint management, experiment lineage, reproducibility, collaboration, SDK/API.